

Sales E-commerce Data Cleaning, Analysis & Visualization

Data Cleaning & Visualization Project
Internship Project

Tools Used	Python, Pandas, NumPy, Matplotlib, Seaborn, Streamlit
Dataset	Online Retail Transaction Dataset (UK-based e-commerce)
Raw Records	541,909 rows × 8 columns
Cleaned Records	524,878 rows × 9 columns
Transaction Period	2010-12-01 to 2011-12-09
Report Date	August 2026

Note: Dataset source metadata (original provider/platform) was not embedded in the uploaded files; this report treats the data strictly as provided.

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1. Executive Summary

This report documents an internship data-cleaning and visualization project performed on a UK-based e-commerce transaction dataset. The raw dataset contains **541,909 transaction line items** across **8 columns**, covering the period from **2010-12-01** to **2011-12-09**. Each row represents one product line within a customer invoice, capturing invoice number, product (stock code and description), quantity, unit price, invoice date, customer ID, and country.

The primary objective was to clean and validate the raw data, then perform exploratory analysis to surface business-relevant patterns in sales, products, customers, and geography. The main data-quality issues identified in the raw dataset were: **24.93% missing Customer IDs** (135,080 rows), **1,454 missing product descriptions**, **5,268 exact duplicate rows** (0.97%), **9,288 cancelled-order line items** (invoice numbers prefixed with 'C'), and **10,624 rows with negative quantities**. After cleaning, the dataset was reduced to **524,878 valid transaction rows** (a reduction of 3.14%), with zero duplicate rows, zero negative quantities, and zero non-positive prices remaining.

Based on the cleaned data, total recorded revenue across the observed period is **£10,642,110.80** from **19,960 unique orders** placed by **4,338 identified customers**, with an average order value of **£533.17**. **84.59%** of total revenue originates from the United Kingdom, with the remaining sales spread across 37 other countries. The highest-revenue month in the dataset was **2011-11** (£1,503,866.78), while the lowest was **2011-02** (£522,545.56).

2. Project Objectives

This project was carried out with the following objectives:

- Clean and preprocess the raw e-commerce transaction dataset
- Identify and handle missing values across all columns
- Identify and remove duplicate transaction records
- Detect invalid data, including negative quantities, non-positive prices, and cancelled-order entries
- Identify potential outliers in quantity and sales-value fields using statistical methods
- Perform exploratory data analysis (EDA) to understand sales, product, customer, and geographic patterns
- Create meaningful, labeled visualizations to communicate findings
- Extract actionable business insights and recommendations from the cleaned data

3. Dataset Overview

3.1 Raw Dataset

The raw dataset (**data.csv**) contains **541,909 rows** and **8 columns**. Each row represents a single product line item within a customer invoice.

Column	Data Type	Description
InvoiceNo	Text	Unique invoice identifier; values prefixed with 'C' denote cancelled orders
StockCode	Text	Unique product/item code
Description	Text	Product name/description
Quantity	Integer	Number of units purchased per line item
InvoiceDate	Text/Date	Date and time the invoice was generated
UnitPrice	Decimal	Price per unit, in GBP (£)
CustomerID	Decimal	Unique customer identifier (missing for guest/unregistered transactions)
Country	Text	Country of the customer

Numerical columns: Quantity, UnitPrice, CustomerID (identifier, not a measure). **Categorical / text columns:** InvoiceNo, StockCode, Description, Country. **Date column:** InvoiceDate, spanning **2010-12-01** to **2011-12-09**.

3.2 Cleaned Dataset

The cleaned dataset (**cleaned_data.csv**) contains **524,878 rows** and **9 columns**. It retains all original columns and adds one derived column, **total_sales** (calculated as $\text{Quantity} \times \text{UnitPrice}$), which was not present in the raw file.

Metric	Raw Dataset	Cleaned Dataset
Rows	541,909	524,878
Columns	8	9
Unique Invoices	25,900	19,960

Metric	Raw Dataset	Cleaned Dataset
Unique Products	4,070	3,922
Unique Customers	4,372	4,338
Unique Countries	38	38
Duplicate Rows	5,268	0

4. Data Quality Analysis

This section quantifies data-quality issues found in the raw dataset prior to cleaning.

4.1 Missing Values

Column	Missing Count	Missing %
Description	1,454	0.27%
CustomerID	135,080	24.93%

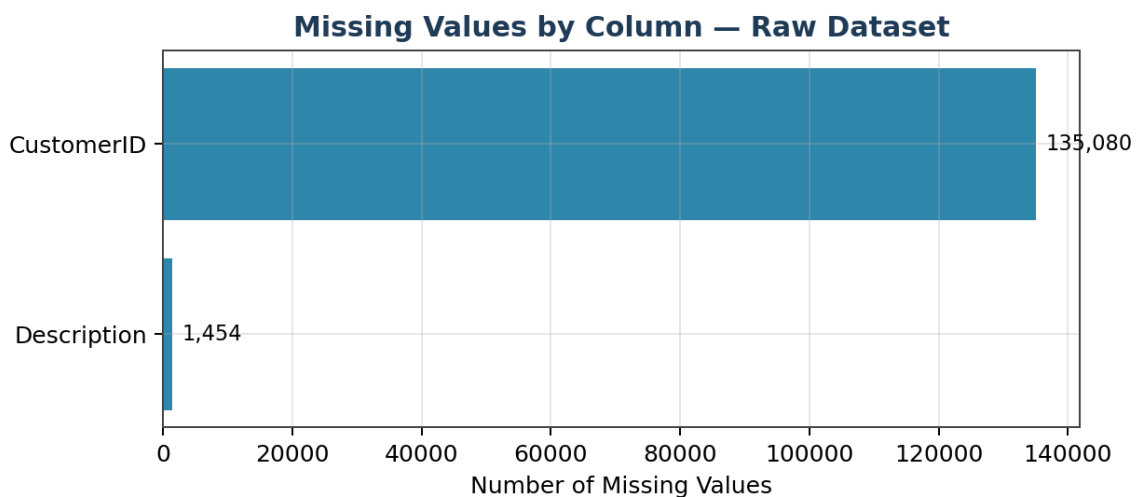


Figure 4.1 — Only Description and CustomerID contain missing values in the raw dataset. All other columns are fully populated.

CustomerID is missing in **24.93%** of rows, most likely corresponding to guest checkouts or unregistered transactions where no customer account was linked. This cannot be inferred or reconstructed from the available data, so these rows were retained but excluded from customer-level analysis.

4.2 Duplicate Records

The raw dataset contains **5,268 exact duplicate rows** (0.97% of all records), where every column value is identical to another row. These are consistent with repeated scanning/logging of the same transaction line and were removed during cleaning.

4.3 Invalid / Inconsistent Values

Issue	Row Count	% of Raw Data
Negative Quantity (returns/adjustments)	10,624	1.96%
Negative Unit Price	2	0.0%
Zero Unit Price	2,515	0.46%
Cancelled Invoices (prefix 'C')	9,288	1.71%

Negative quantities correspond almost entirely to cancelled invoices (InvoiceNo beginning with 'C'), which represent product returns rather than sales. Rows with zero or negative unit price represent non-revenue transactions (e.g., free samples, write-offs, or data-entry adjustments) that would distort revenue calculations if included.

4.4 Potential Outliers

Using the Interquartile Range (IQR) method on the cleaned dataset's **total_sales** field (line-item revenue), values above **£38.40** were flagged as statistical outliers. This identified **42,624 line items** (8.12% of cleaned rows) as high-value outliers. Similarly, on the **quantity** field, values above **26.0** units were flagged, affecting **27,111 line items**. These are legitimate bulk/wholesale orders rather than data errors, and were therefore **retained** in the cleaned dataset — see Section 5.6 for the rationale.

Outlier Detection — Line-Item Sales Value (Boxplot, IQR Method)

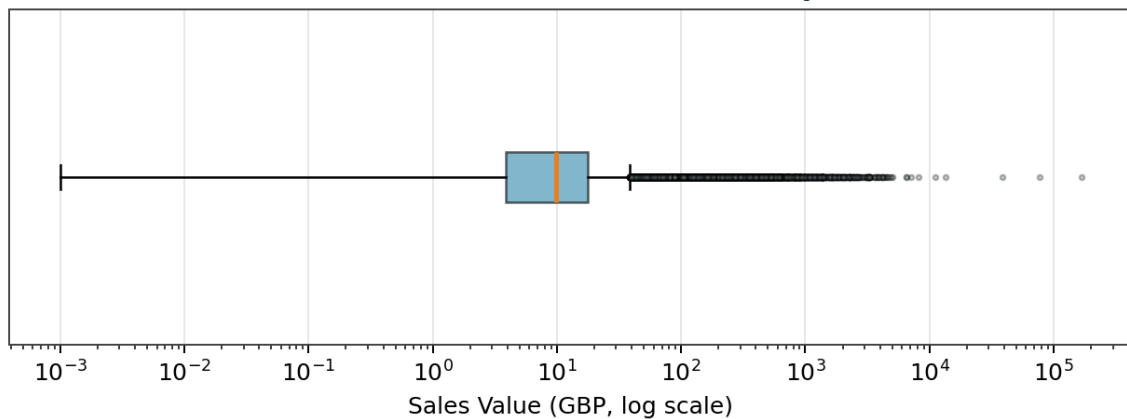


Figure 4.2 — Boxplot of line-item sales value (log scale) showing the spread of high-value outliers typical of wholesale/bulk transactions.

5. Data Cleaning & Preprocessing

The following steps were applied to transform the raw dataset into the cleaned dataset:

5.1 Missing-Value Treatment

- **CustomerID** (135,080 missing, 24.93%): retained as null. These represent guest/unregistered transactions and cannot be reliably imputed; they are excluded only from customer-level aggregations (e.g., unique customer counts, per-customer spend).
- **Description** (1,454 missing): rows with a missing product description were removed, since the product cannot be identified for product-level analysis.

5.2 Duplicate Removal

All 5,268 exact duplicate rows were removed, keeping the first occurrence of each. The cleaned dataset contains zero duplicate rows.

5.3 Data-Type Conversion

- InvoiceDate was converted from text to a proper datetime type to enable time-based aggregation (monthly/yearly trends, day-of-week analysis).
- Quantity and UnitPrice were validated as numeric (integer / float) types.

5.4 Invalid-Value Handling

- Cancelled-order line items (InvoiceNo prefixed with 'C') and rows with negative quantity were removed, as they represent returns/reversals rather than completed sales and would understate revenue if summed together with sales.
- Rows with zero or negative UnitPrice were removed, as they do not represent valid revenue-generating transactions.

5.5 Text / Category Standardization

Column names were standardized to lowercase for consistency (e.g., InvoiceNo → invoiceno). Product descriptions and country names were retained as provided in the source data; no further re-categorization was applied, as doing so without a validated mapping would risk introducing assumptions not supported by the data.

5.6 Outlier Treatment

Statistical outliers in quantity and sales value (see Section 4.4) were identified but **not removed**. Review of the top outlier records shows they correspond to bulk/wholesale orders and postage/manual adjustment line items rather than data-entry errors. Removing them would understate true business volume, so they were retained; analysts should be aware of their influence on mean-based statistics (see Section 6).

5.7 Derived Column

A **total_sales** column was added, calculated as $\text{Quantity} \times \text{UnitPrice}$, to support revenue-based analysis throughout this report.

5.8 Cleaning Summary

Metric	Value
Raw row count	541,909
Rows removed during cleaning	17,031
% of data removed	3.14%
Cleaned row count	524,878

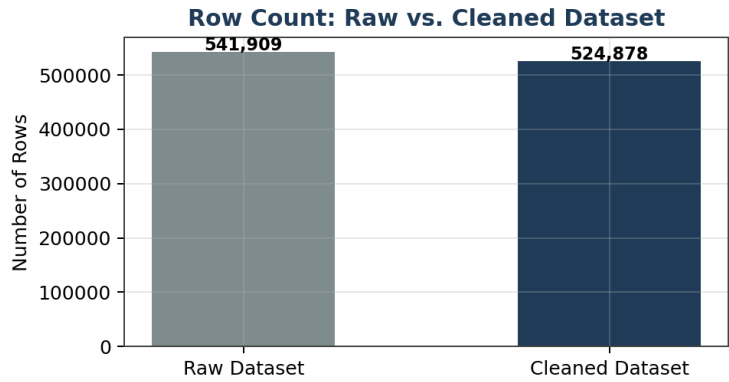


Figure 5.1 — Row count before and after the cleaning process.

6. Exploratory Data Analysis

All figures in this section are calculated from the cleaned dataset.

6.1 Key Business Metrics

Metric	Value
Total Revenue	£10,642,110.80
Total Orders (unique invoices)	19,960
Total Line Items	524,878
Unique Customers	4,338
Average Order Value	£533.17
Total Units Sold (Quantity)	5,572,420
Average Orders per Customer	4.27
Median Orders per Customer	2.0
Average Spend per Customer	£2,048.69

Note: Average Order Value and Average Orders per Customer are influenced by a small number of very large wholesale orders (see Section 4.4 on outliers); the median orders-per-customer figure above is provided as a more representative measure of typical customer behavior.

6.2 Sales by Month and Year

Yearly revenue: **2010**: £821,452.73 (partial year — data begins December 2010); **2011**: £9,820,658.07. The highest-grossing month was **2011-11** at £1,503,866.78, and the lowest-grossing complete month was **2011-02** at £522,545.56. Note that the final month in the dataset (2011-12) is a partial month, only including data through 2011-12-09, and should not be compared directly to full months.

6.3 Sales by Country

Country	Total Sales
United Kingdom	£9,001,744.09
Netherlands	£285,446.34
EIRE	£283,140.52
Germany	£228,678.40
France	£209,625.37
Australia	£138,453.81
Spain	£61,558.56
Switzerland	£57,067.60
Belgium	£41,196.34
Sweden	£38,367.83

The United Kingdom accounts for **84.59%** of total revenue (£9,001,744.09), reflecting the dataset's origin as a UK-based retailer. The remaining 15.41% is distributed across 37 other countries, led by the Netherlands, Ireland (EIRE), and Germany.

6.4 Top-Selling Products

By revenue:

Product Description	Total Revenue
DOTCOM POSTAGE	£206,248.77
REGENCY CAKESTAND 3 TIER	£174,156.54
PAPER CRAFT , LITTLE BIRDIE	£168,469.60
WHITE HANGING HEART T-LIGHT HOLDER	£106,236.72
PARTY BUNTING	£99,445.23
JUMBO BAG RED RETROSPOT	£94,159.81
MEDIUM CERAMIC TOP STORAGE JAR	£81,700.92
POSTAGE	£78,101.88
Manual	£77,752.82
RABBIT NIGHT LIGHT	£66,870.03

Note: 'DOTCOM POSTAGE', 'POSTAGE', and 'Manual' are administrative/shipping line items rather than physical products, and their appearance in the top-revenue list should be interpreted accordingly.

By quantity sold:

Product Description	Units Sold
PAPER CRAFT , LITTLE BIRDIE	80,995
MEDIUM CERAMIC TOP STORAGE JAR	78,033
WORLD WAR 2 GLIDERS ASSTD DESIGNS	54,951
JUMBO BAG RED RETROSPOT	48,371
WHITE HANGING HEART T-LIGHT HOLDER	37,872
POPCORN HOLDER	36,749
PACK OF 72 RETROSPOT CAKE CASES	36,396
ASSORTED COLOUR BIRD ORNAMENT	36,362
RABBIT NIGHT LIGHT	30,739
MINI PAINT SET VINTAGE	26,633

6.5 Customer Purchasing Patterns

Among the 4,338 identified customers, the average customer placed **4.27 orders** (median: 2.0), spending an average of **£2,048.69** over the observed period. Spending is highly concentrated: the top 10 customers by total spend are shown below.

Customer ID	Total Spend
Customer #14646	£280,206.02
Customer #18102	£259,657.30
Customer #17450	£194,390.79
Customer #16446	£168,472.50
Customer #14911	£143,711.17
Customer #12415	£124,914.53
Customer #14156	£117,210.08
Customer #17511	£91,062.38
Customer #16029	£80,850.84
Customer #12346	£77,183.60

6.6 Sales by Day of Week

Day of Week	Total Sales
Friday	£1,837,470.49
Monday	£1,775,782.07
Sunday	£806,790.78
Thursday	£2,199,292.57
Tuesday	£2,175,700.51
Wednesday	£1,847,074.38

No transactions are recorded on Saturdays in this dataset, indicating the business did not process or log orders on that day during the observed period.

7. Data Visualization

This section presents the key visualizations generated from the cleaned dataset.

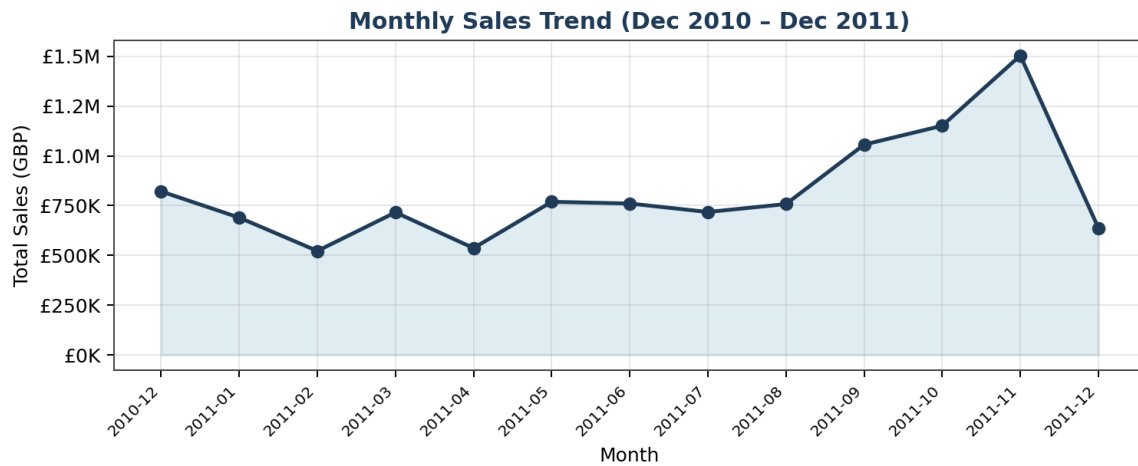


Figure 7.1 — Monthly sales trend. Revenue rises through autumn 2011, peaking in November 2011 ahead of the holiday season, before the partial final month (Dec 2011) shows a lower total.

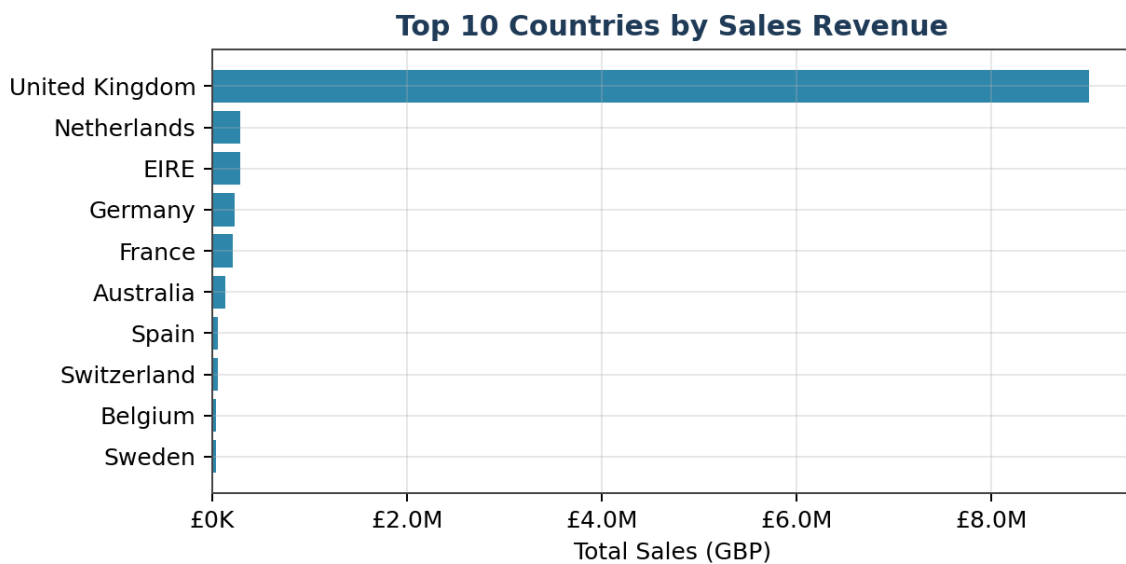


Figure 7.2 — Top 10 countries by total sales revenue. The United Kingdom dominates, consistent with the dataset's origin.

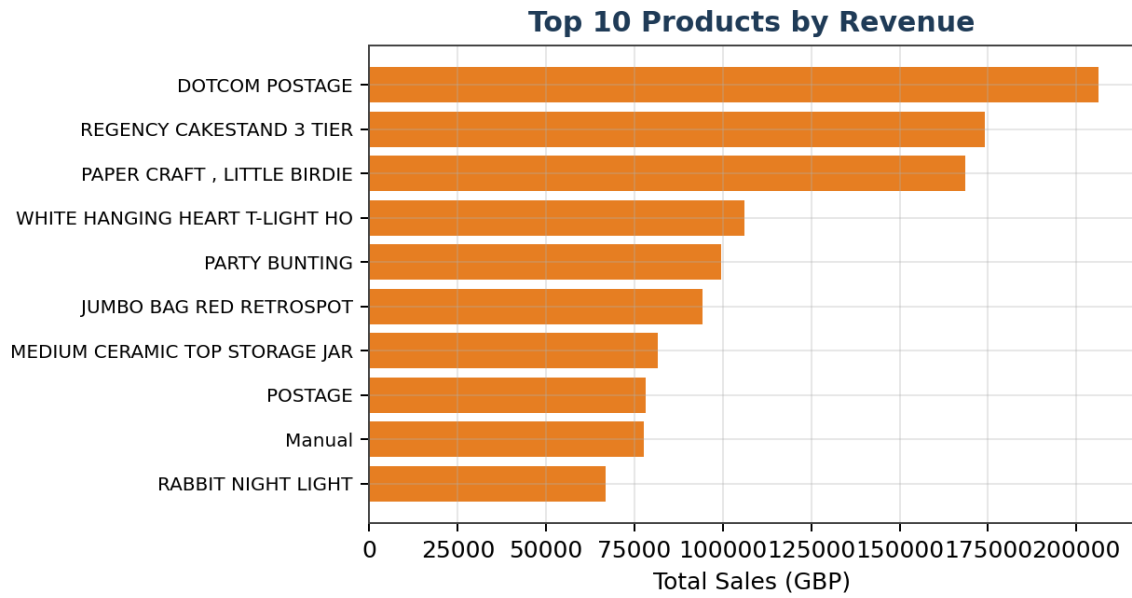


Figure 7.3 — Top 10 products by total revenue generated.

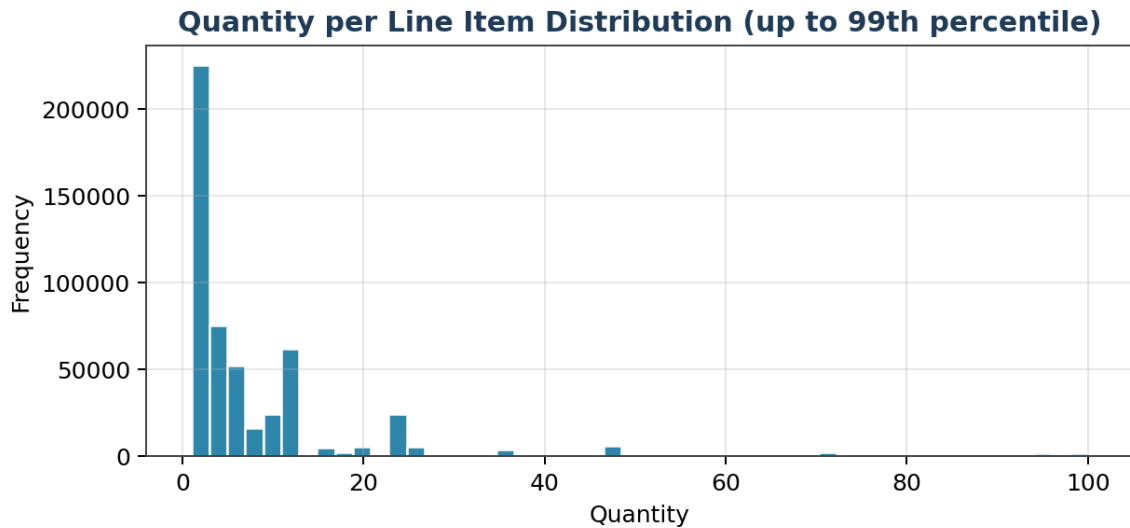


Figure 7.4 — Distribution of quantity per line item (values above the 99th percentile excluded for chart readability). Most orders involve small quantities, with a long right tail of bulk orders.

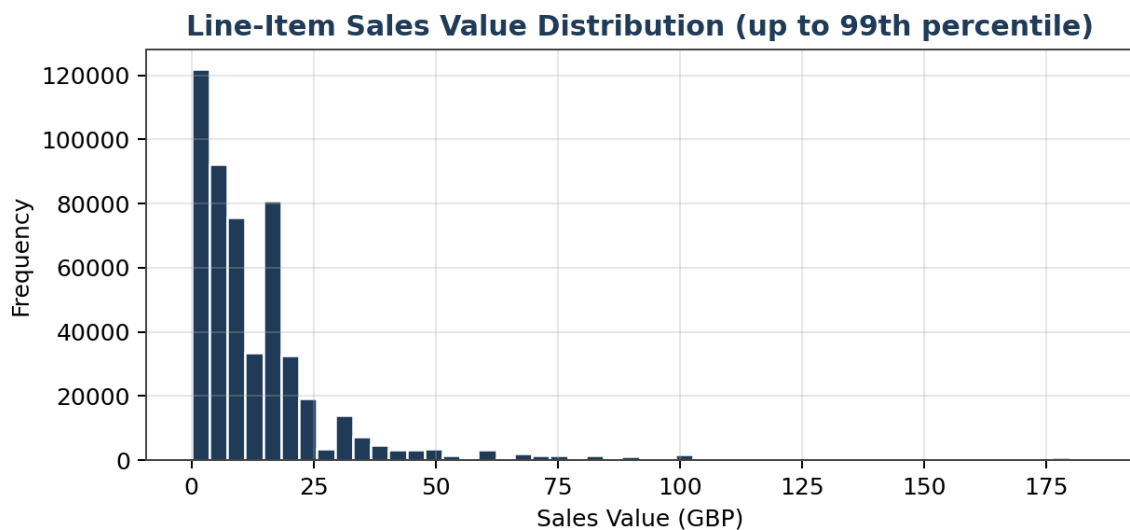


Figure 7.5 — Distribution of line-item sales value (values above the 99th percentile excluded for chart readability).

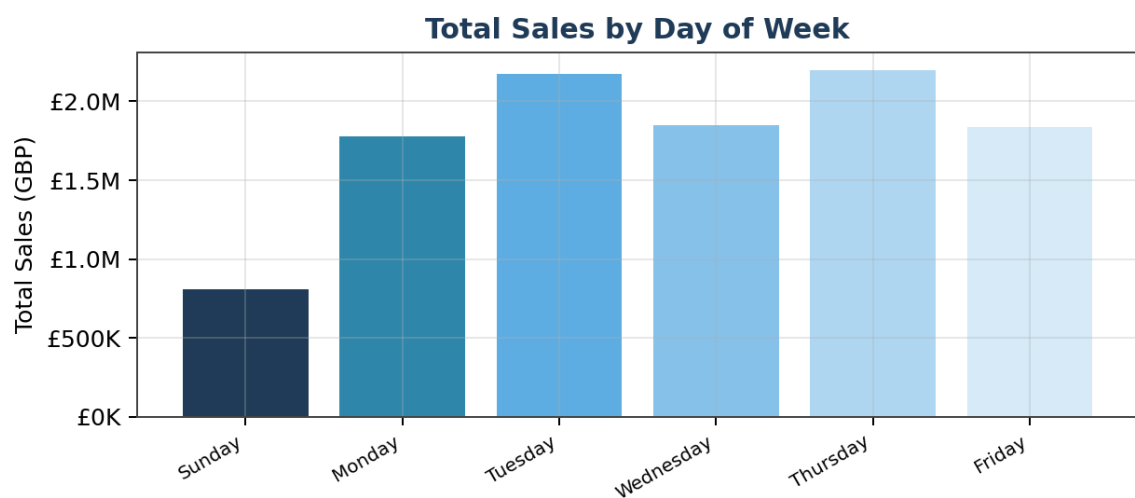


Figure 7.6 — Total sales by day of week. No Saturday transactions are present in the dataset.

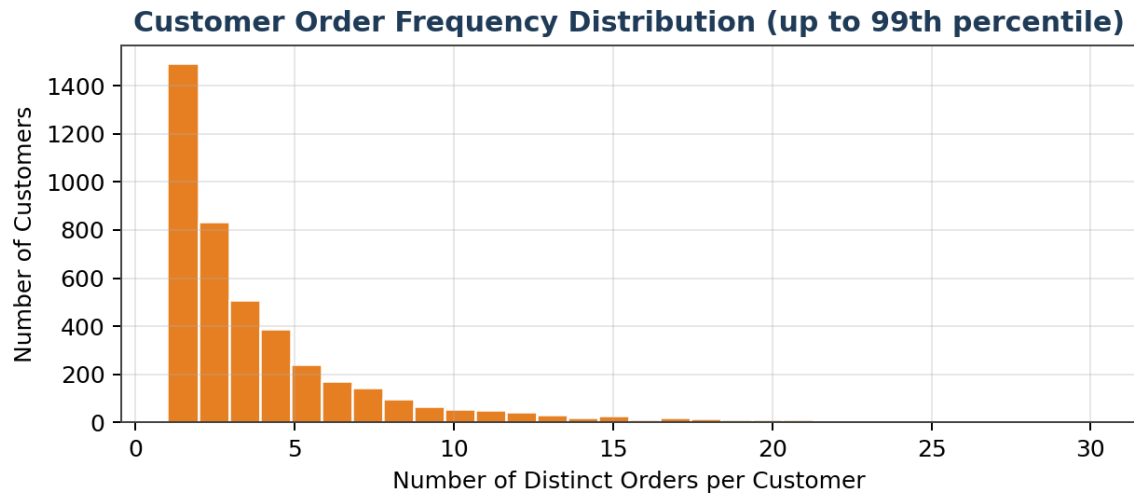


Figure 7.7 — Distribution of the number of distinct orders placed per customer (values above the 99th percentile excluded for readability). Most customers place only a few orders.

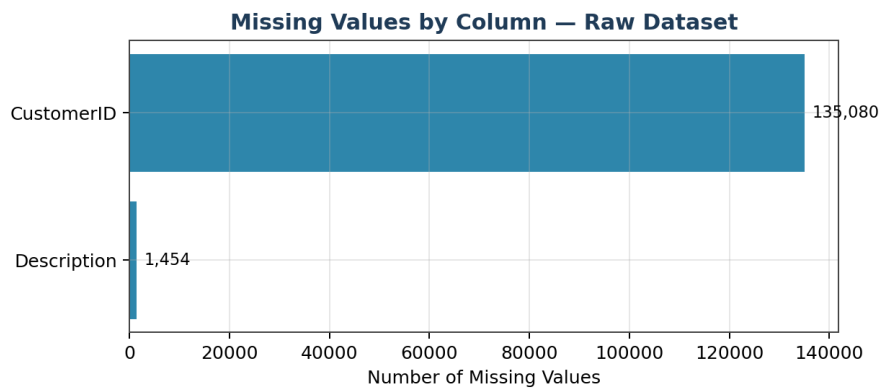


Figure 7.8 — Missing-value counts by column in the raw dataset (repeated here for reference).

8. Business Insights

- **Revenue is heavily concentrated in the UK:** 84.59% of total revenue comes from the United Kingdom, making it the dominant market by a wide margin over the next-largest market (Netherlands).
- **Sales are seasonal:** monthly revenue trends upward from August through November 2011, peaking in November (£1,503,866.78) — consistent with pre-holiday purchasing.
- **Revenue is driven by a small set of high-value orders:** 8.12% of line items (those above £38.40) are statistical outliers, indicating a mix of retail and wholesale/bulk buying behavior in the customer base.
- **Customer spend is concentrated among a small group:** the top 10 customers alone account for £1,537,659.21 in total spend, out of 4,338 customers overall — highlighting the importance of key-account relationships.
- **Most customers are infrequent buyers:** the median customer places only 2.0 orders, while the mean (4.27) is pulled upward by a smaller group of repeat buyers.
- **Administrative charges appear alongside product sales:** 'POSTAGE', 'DOTCOM POSTAGE', and 'Manual' entries rank among the top revenue line items; these are not physical products and should be excluded from any product-performance ranking used for merchandising decisions.
- **Data-quality issues could materially affect naive analysis:** the raw dataset's 24.93% missing CustomerID rate would understate customer counts if not accounted for, and its unfiltered negative-quantity/cancelled-order rows would understate net revenue if summed directly without cleaning.
- **No Saturday activity is recorded:** this is either a genuine business pattern (no weekend order processing) or a data-collection gap; it cannot be determined which from the data alone.

9. Key Findings

- Raw dataset: 541,909 rows, 8 columns; cleaned dataset: 524,878 rows (3.14% removed).
- 24.93% of raw rows had missing CustomerID; 5,268 exact duplicate rows were found and removed.
- 9,288 rows were cancelled orders (negative quantity) and were excluded from revenue calculations.
- Total cleaned-data revenue: £10,642,110.80 across 19,960 orders and 4,338 customers.
- Average order value: £533.17; average spend per customer: £2,048.69.
- United Kingdom contributes 84.59% of total revenue.
- Peak sales month: 2011-11 (£1,503,866.78).
- 8.12% of line items are statistical high-value outliers, reflecting bulk/wholesale orders.

10. Recommendations

- Capture CustomerID consistently at checkout to reduce the current ~25% missing-ID rate, enabling more complete customer-level analysis and personalization.
- Separate administrative line items (postage, manual adjustments) from product-level sales reporting so that merchandising and inventory decisions are based on actual product performance.
- Investigate the concentration of revenue among a small number of high-spend customers and consider a dedicated key-account or loyalty program to protect and grow this segment.
- Given the strong seasonal peak around November, plan inventory and staffing ahead of the autumn/holiday period to capture demand and avoid stockouts.
- Review order-entry and cancellation workflows so that returns/cancellations are logged with a consistent, queryable flag rather than relying solely on an invoice-number prefix convention.

- Confirm whether the absence of Saturday transactions reflects an actual operating schedule or a data-logging gap, since this materially affects any day-of-week demand planning.

11. Conclusion

This project cleaned a raw e-commerce transaction dataset of 541,909 rows down to a validated, analysis-ready dataset of 524,878 rows by removing exact duplicates, cancelled/negative-quantity transactions, and non-positive-price entries, while preserving legitimate high-value orders. This directly improved the reliability of downstream revenue and customer metrics, which would otherwise have been distorted by returns and duplicate records.

Exploratory analysis and visualization revealed clear patterns in geography (UK dominance), seasonality (a Q4 sales peak), and customer concentration (a small group of high-value customers), which would not have been readily apparent from the raw tabular data alone. The visualizations in Section 7 translate these statistics into a form suitable for non-technical business stakeholders.

Overall, the project demonstrates an end-to-end data-cleaning and analysis workflow — from raw-data audit through cleaning, exploratory analysis, and visualization — appropriate for an internship-level data analytics deliverable.

12. Tools & Technologies

- **Python** — core programming language for data processing
- **Pandas** — data cleaning, transformation, and aggregation
- **NumPy** — numerical computation
- **Matplotlib** and **Seaborn** — data visualization
- **Streamlit** — (project tooling, as specified for this internship)
- **Git/GitHub** — version control

13. Appendix

13.1 Final Dataset Dimensions

Item	Value
Final cleaned rows	524,878
Final cleaned columns	9
Columns	invoiceno, stockcode, description, quantity, invoicedate, unitprice, customerid, country, total_sales

13.2 Cleaning Statistics Summary

Metric	Value
Raw rows	541,909
Cleaned rows	524,878
Rows removed	17,031 (3.14%)
Duplicate rows removed	5,268
Cancelled/negative-quantity rows removed	10,624
Non-positive price rows removed	2,517
Rows with missing Description removed	1,454
Missing CustomerID rows retained (not imputed)	132,186

13.3 Descriptive Statistics — Cleaned Dataset

Field	Mean	Median	Min	Max
Quantity	10.62	4.0	1.0	80995.0
Unit Price (£)	3.92	2.08	0.0	13541.33
Line Total Sales (£)	20.28	9.92	0.0	168469.6

All figures in this report were computed directly from the uploaded `data.csv` (raw) and `cleaned_data.csv` (cleaned) files. No statistics were estimated or fabricated. Where information could not be determined from the data (e.g., original dataset source/provider), this has been explicitly stated rather than assumed.